Lesson 1-4 • Continuity at a Point

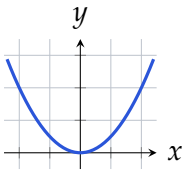
Notes (filled in)

Objective: identify the types of discontinuities and prove continuity with the continuity test.

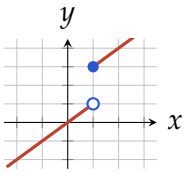
What does it mean to be continuous?

A graph is continuous if you can trace it without lifting your pencil.

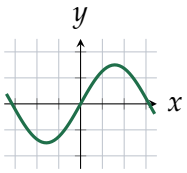
Are these graphs continuous or discontinuous?



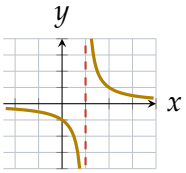
continuous



discontinuous

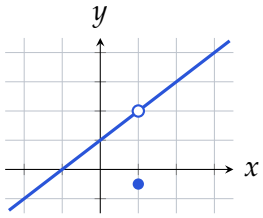


continuous



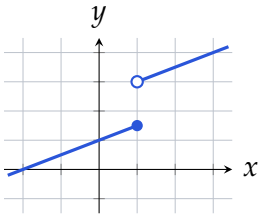
discontinuous

Types of discontinuities



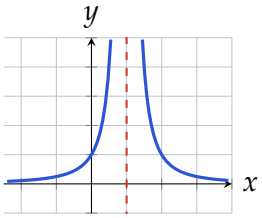
Removable (hole)

the limit exists but is not equal to $f(c)$



Jump

the one-sided limits are different



Infinite (VA)

the function goes to $\pm\infty$

An **endpoint** (like the start of $y = \sqrt{x}$ at $x = 0$) only needs the one-sided limit from the inside of the domain.

Continuity test

f is continuous at $x = c$ when

$$\underbrace{\lim_{x \to c^-} f(x)}_{\text{left}} = \underbrace{\lim_{x \to c^+} f(x)}_{\text{right}} = \underbrace{f(c)}_{\text{actual height}}$$

Example 1

$$h(x) = \begin{cases} 2x - 1, & x \leq 3 \\ x^2 - 4, & x > 3 \end{cases}$$

Prove that h is continuous at $x = 3$.

Left: $\lim_{x \to 3^-} h(x) = 2(3) - 1 = 5$. Right: $\lim_{x \to 3^+} h(x) = 3^2 - 4 = 5$.
Height: $h(3) = 2(3) - 1 = 5$.
$5 = 5 = 5$, so h is continuous at $x = 3$ because it passes the continuity test.

Example 2	For what value of k is g continuous at $x = 1$? $g(x) = \begin{cases} 3x^2 - k, & x \le 1 \\ kx + 5, & x > 1 \end{cases}$ Set left $=$ right at $x = 1$: $3(1)^2 - k = k(1) + 5$, so $3 - k = k + 5$, $-2 = 2k$, $k = -1$. Check: left $3 - (-1) = 4$, right $-1 + 5 = 4$, $g(1) = 4$. Continuous.
Summary	Continuous means no hole, no jump and no asymptote at the point. To prove it, show left limit $=$ right limit $= f(c)$. To find a constant, set the two pieces equal at the suspect point and solve.